Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam C · Form 1: Connecting ideas

Multiple Choice Questions: Select the best answer. (1 point each)

1. A cart velocity graph encloses 6 m above and 6 m below the time axis. Which claim is consistent with the graph?

(A) Maximum displacement must be 12 m.

(B) It returns to the start after traveling 12 m.

(C) Mean speed is zero.

(D) It must end at rest.

2. A cart is currently at x = -8 m with v = -8 m/s and a = -1 m/s². Which statement correctly describes its motion at this instant?

(A) The object is currently moving to the right.

(B) The x–t slope is zero and the object is at rest.

(C) A negative velocity means a positive x–t slope.

(D) It moves left and speeds up.

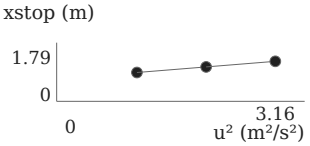
3. A cart brakes immediately at a constant magnitude of 2 m/s². Read its stopping coordinate along the track. u² (m²/s²) -> xstop (m): (1.16, 1.29); (2.16, 1.54); (3.16, 1.79). Which interpretation of the experiment is valid?

(A) The slope is the first vertical reading.

(B) A reaction delay adds distance proportional to u.

(C) Horizontal interval alone determines vertical change without a model.

(D) Exchanging axes always preserves the numerical slope.



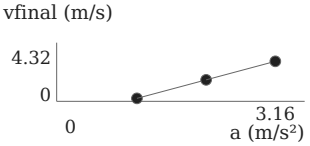
4. A cart initially moving at -2 m/s receives a constant acceleration lasting 2 s each trial. a (m/s²) -> vfinal (m/s): (1.16, 0.32); (2.16, 2.32); (3.16, 4.32). By what factor does the first-to-third change exceed the first-to-second change?

(A) 2

(B) 1.2

(C) 2.8

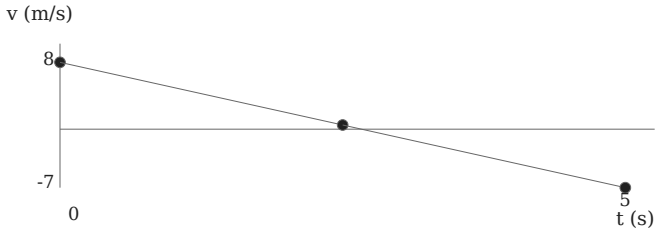
(D) 3.6



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A sensor records a cart with x0 = -4 m and v0 = +8 m/s. It has constant a = -3 m/s² for 5 s. Right is positive.

a. Describe the shapes and initial values of x–t, v–t, and a–t. **(1 pt)**



b. Find final position, displacement, and distance; explain any reversal. **(3 pt)**

6. A sensor records downward position for a weight dropped from rest. Use g=10 m/s². t² (s²) -> y (m): (1.16, 7.8); (2.16, 12.8); (3.16, 17.8).

a. Build the model for drop sensor from physics and predict both slope and intercept. **(2 pt)**

b. Exchange the axes. Relate the new slope and its units to the original slope. **(2 pt)**

7. An object with constant acceleration a has initial and final velocities u and v. Eliminate time to derive $v^2 = u^2 + 2a\Delta x$. **(3 pt)**